# Physics CS

# Code: EBA1110


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# Course Content

- Lecture 1: Units and Vectors.
- Lecture 2: Newton Laws of motion &Gravitational Force, weight and mass
- Lecture 3: Work, Kinetic and Potential Energy, Work Energy theorem& Conservation law of Energy

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- Lecture 4: Oscillatory motion , simple and Harmonic Motion, Introduction to mechanical waves properties , and Travelling wave
- Lecture 5: Sound wave Power, Intensity, speed in solid , liquid or gas , Sound level calculation
- Lecture 6: Superposition of waves.

# Course Content

- Lecture 7: capacitors.
- Lecture 8: Calculation of the Electric Current, Resistance, Ohm's law, Power Calculation.
- Lecture 9: Electromotive force , Internal Resistance , Resistance in Series and Parallel, Kirchhoff law and RC Circuit
- Lecture 10: Magnetic Force
- Lecture 11: Biot Savart Law, Ampere Law and their applications. Magnetic force for two parallel conductors.Explanation of Faraday Law of Induction

# LECTURE 1
# Units and Vectors.

- Why use units

This table is 1000 long.

This table is 1 long.



Mohamed Salah is 175 tall

Mohamed Salah is

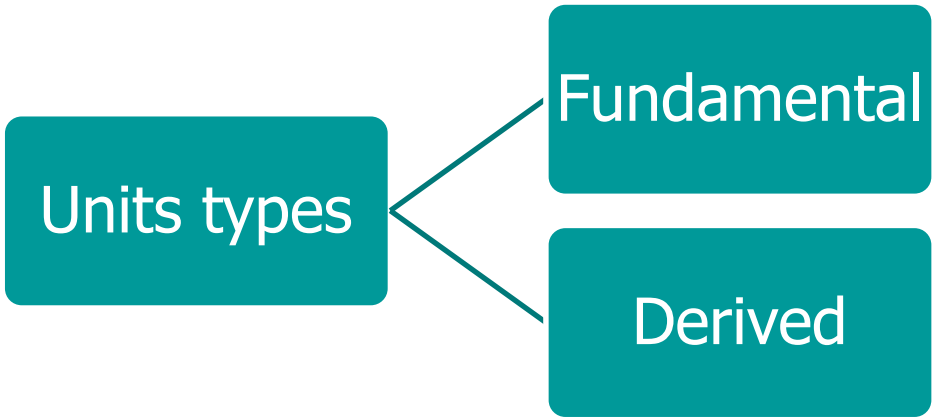
17500 tall



## Units:

It is the description of any physical quantity to know what its mean. Because the change in this unit will be change the meaning of this value.

الوحدات : هي عملية توصيفية لاي قيمة فيزيائية لمعرفة معني الرقم.
و ذلك لان التغيرفي نوع الوحدات يغير المعني تماماً للقيمة المعطاه.

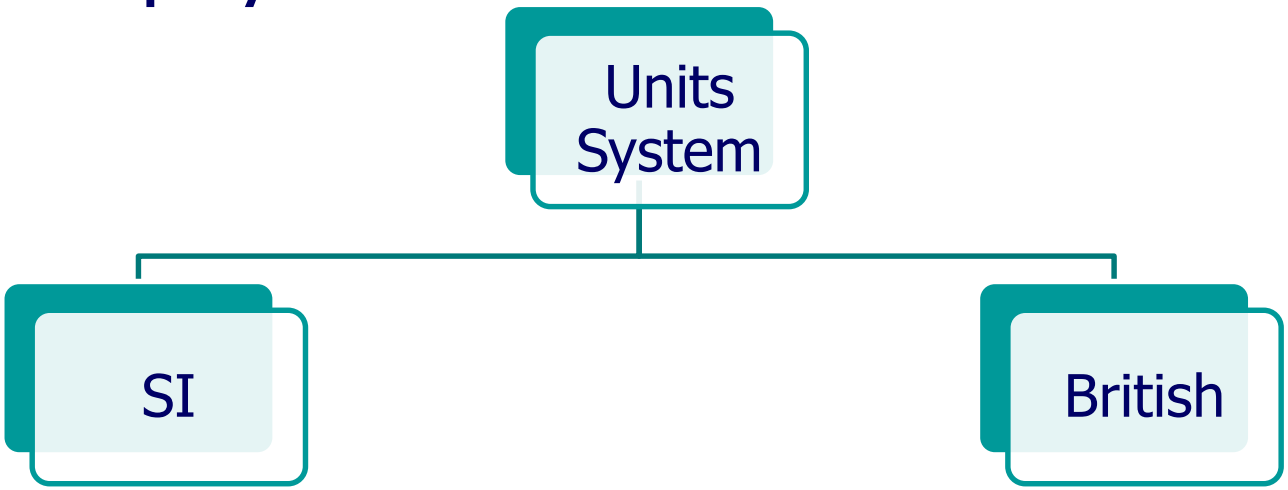



Length, Mass, Time, Temperature, Current and Luminous intensity

Velocity, Energy, Density ,…ect.

# Standard units

- Length, mass, and time are three fundamental quantities of physics.




| | | |
|---|---|---|
| ➢Length | m | ft |
| ➢Mass | Kg | Ib |
| ➢Time | S | S |

The International System (SI for System International) is the most widely used system of units.

# Table 1 shows some larger and smaller units for the fundamental quantities.

**Table 1.1 Some Units of Length, Mass, and Time**

| Length | Mass | Time |
|---|---|---|
| 1 nanometer = 1 nm = $10^{-9}$ m<br>(a few times the size of the largest atom) | 1 microgram = 1 $\mu$g = $10^{-6}$ g = $10^{-9}$ kg<br>(mass of a very small dust particle) | 1 nanosecond = 1 ns = $10^{-9}$ s<br>(time for light to travel 0.3 m) |
| 1 micrometer = 1 $\mu$m = $10^{-6}$ m<br>(size of some bacteria and living cells) | 1 milligram = 1 mg = $10^{-3}$ g = $10^{-6}$ kg<br>(mass of a grain of salt) | 1 microsecond = 1 $\mu$s = $10^{-6}$ s<br>(time for space station to move 8 mm) |
| 1 millimeter = 1 mm = $10^{-3}$ m<br>(diameter of the point of a ballpoint pen) | 1 gram = 1 g = $10^{-3}$ kg<br>(mass of a paper clip) | 1 millisecond = 1 ms = $10^{-3}$ s<br>(time for sound to travel 0.35 m) |
| 1 centimeter = 1 cm = $10^{-2}$ m<br>(diameter of your little finger) | | |
| 1 kilometer = 1 km = $10^{3}$ m<br>(a 10-minute walk) | | |

# Vectors and scalars

- Scaler Quantity

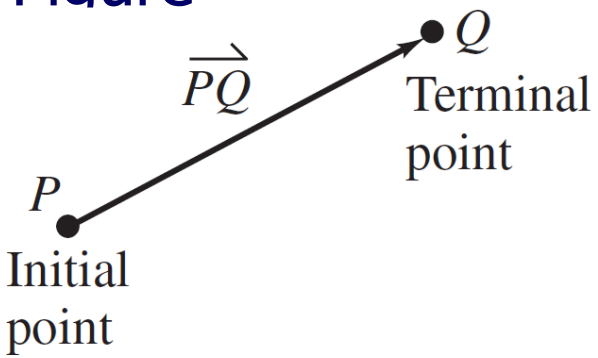
Many quantities in geometry and physics can be represented by a single real number.

Example : time, temperature, mass, money.

- Vector Quantity

Other quantities cannot be completely characterized by a single real number but involve both *magnitude* and *direction* . To represent such a quantity, you can use a **directed line segment,** as shown in Figure

Example : force , velocity , acceleration.



## Some Properties of Vectors

In this section, we shall investigate general properties of vectors representing physical quantities. We also discuss how to add and subtract vectors using both algebraic and geometric methods.

### Equality of Two Vectors

For many purposes, two vectors $\vec{A}$ and $\vec{B}$ may be defined to be equal if they have the same magnitude and if they point in the same direction. That is, $\vec{A} = \vec{B}$ only if $A = B$ and if $\vec{A}$ and $\vec{B}$ point in the same direction along parallel lines. For example, all the vectors in Figure 3.5 are equal even though they have different starting points. This property allows us to move a vector to a position parallel to itself in a diagram without affecting the vector.

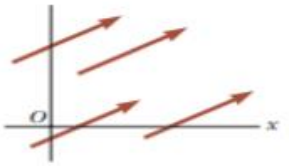



**Figure 3.5** These four vectors are equal because they have equal lengths and point in the same direction.

### Adding Vectors

The rules for adding vectors are conveniently described by a graphical method. To add vector $\vec{B}$ to vector $\vec{A}$, first draw vector $\vec{A}$ on graph paper, with its magnitude represented by a convenient length scale, and then draw vector $\vec{B}$ to the same scale, with its tail starting from the tip of $\vec{A}$, as shown in Figure 3.6. The **resultant vector** $\vec{R} = \vec{A} + \vec{B}$ is the vector drawn from the tail of $\vec{A}$ to the tip of $\vec{B}$.

A geometric construction can also be used to add more than two vectors as shown in Figure 3.7 for the case of four vectors. The resultant vector $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$ is the vector that completes the polygon. In other words, $\vec{R}$ is the vector drawn from the tail of the first vector to the tip of the last vector. This technique for adding vectors is often called the "head to tail method."

When two vectors are added, the sum is independent of the order of the addition. (This fact may seem trivial, but as you will see in Chapter 11, the order is important when vectors are multiplied. Procedures for multiplying vectors are discussed in Chapters 7 and 11.) This property, which can be seen from the geometric construction in Figure 3.8, is known as the **commutative law of addition:**

**Commutative law of addition** ▶ $\vec{A} + \vec{B} = \vec{B} + \vec{A}$

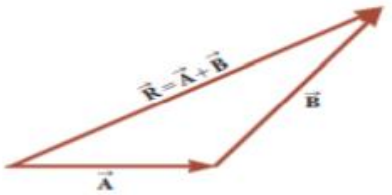



**Figure 3.6** When vector $\vec{B}$ is added to vector $\vec{A}$, the resultant $\vec{R}$ is the vector that runs from the tail of $\vec{A}$ to the tip of $\vec{B}$.

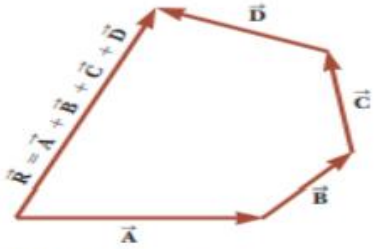



**Figure 3.7** Geometric construction for summing four vectors. The resultant vector $\vec{R}$ is by definition the one that completes the polygon.

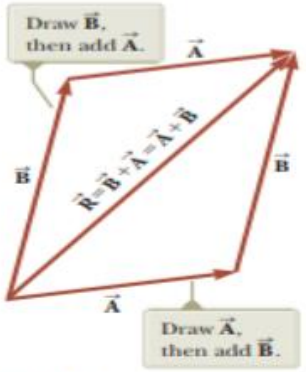



**Figure 3.8** This construction shows that $\vec{A} + \vec{B} = \vec{B} + \vec{A}$ or, in other words, that vector addition is commutative.

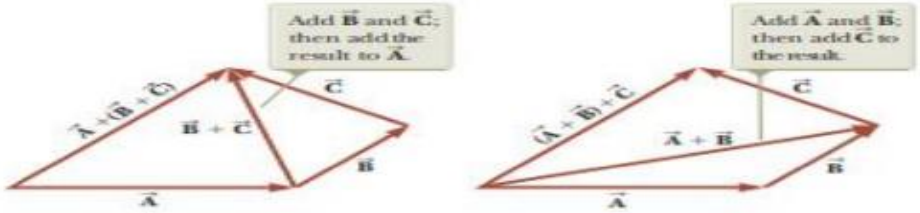



**Figure 3.9** Geometric constructions for verifying the associative law of addition.

When three or more vectors are added, their sum is independent of the way in which the individual vectors are grouped together. A geometric proof of this rule for three vectors is given in Figure 3.9. This property is called the **associative law of addition:**

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} \qquad (3.6)$$

◀ Associative law of addition

In summary, **a vector quantity has both magnitude and direction and also obeys the laws of vector addition** as described in Figures 3.6 to 3.9. When two or more vectors are added together, they must all have the same units and they must all be the same type of quantity. It would be meaningless to add a velocity vector (for example, 60 km/h to the east) to a displacement vector (for example, 200 km to the north) because these vectors represent different physical quantities. The same rule also applies to scalars. For example, it would be meaningless to add time intervals to temperatures.

## Negative of a Vector

The negative of the vector $\vec{A}$ is defined as the vector that when added to $\vec{A}$ gives zero for the vector sum. That is, $\vec{A} + (-\vec{A}) = 0$. The vectors $\vec{A}$ and $-\vec{A}$ have the same magnitude but point in opposite directions.

## Subtracting Vectors

The operation of vector subtraction makes use of the definition of the negative of a vector. We define the operation $\vec{A} - \vec{B}$ as vector $-\vec{B}$ added to vector $\vec{A}$:

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) \qquad (3.7)$$

The geometric construction for subtracting two vectors in this way is illustrated in Figure 3.10a.

Another way of looking at vector subtraction is to notice that the difference $\vec{A} - \vec{B}$ between two vectors $\vec{A}$ and $\vec{B}$ is what you have to add to the second vector to obtain the first. In this case, as Figure 3.10b shows, the vector $\vec{A} - \vec{B}$ points from the tip of the second vector to the tip of the first.

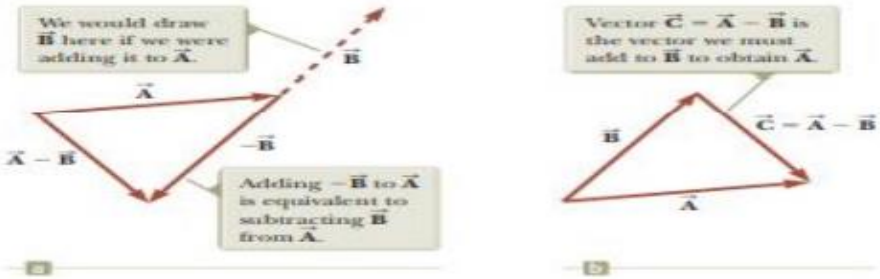
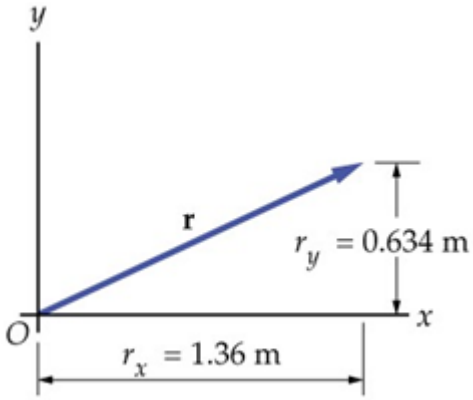



**Figure 3.10** (a) Subtracting vector $\vec{B}$ from vector $\vec{A}$. The vector $-\vec{B}$ is equal in magnitude to vector $\vec{B}$ and points in the opposite direction. (b) A second way of looking at vector subtraction.

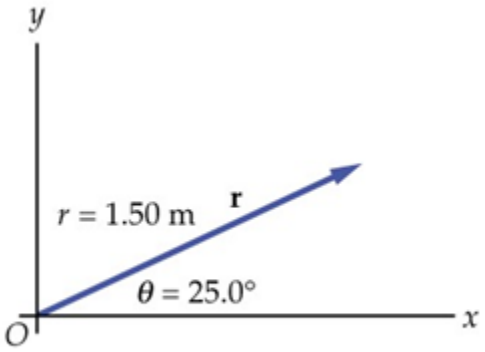
**Vector Representation**

**Cartesian Form**




$(r_x, r_y) = (\ 1.36\ ,\ 0.634\ )$

**Polar Form**




$\vec{A} = |A| \langle \theta^o$

The vector A in general Cartesian form

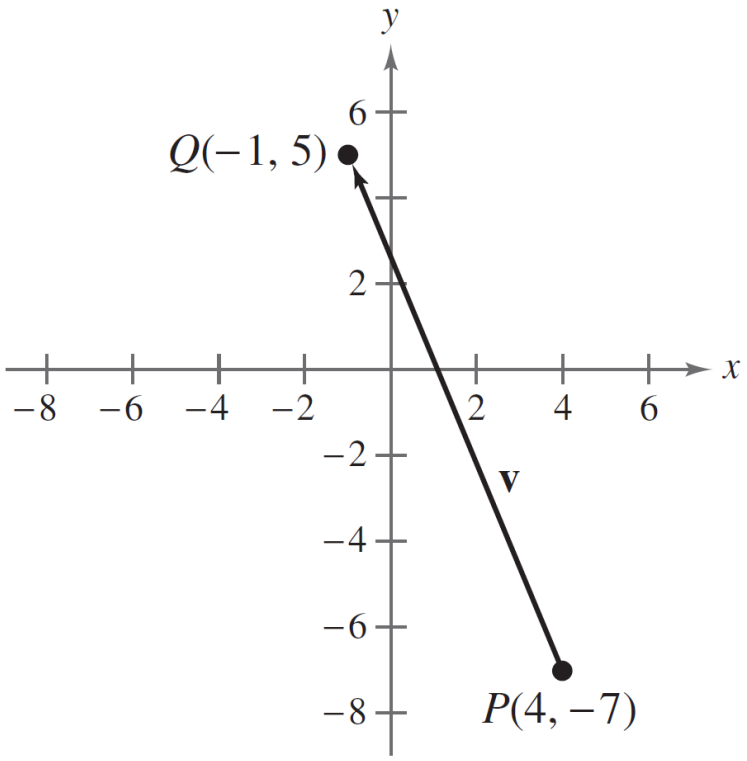
$$\vec{A} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$

The magnitude vector A

$$|A| = \sqrt{x^2 + y^2 + z^2}$$

Example：

Find the component form and magnitude of the vector **v** that has initial point (4, –7) and terminal point (–1, 5).




Solution:

Let P(4, –7)

and Q(–1, 5)

as shown in Figure

Then $\vec{v} = \overrightarrow{PQ}$

$$\vec{v} = (-1-4, 5+7) \qquad \vec{v} = (-5, 12)$$

and the magnitude is $|\vec{v}| = \sqrt{(-5)^2 + (12)^2} = \sqrt{169} = 13$

## Example :

Let **u** be represented by the directed line segment from

$P$ (0, 0) to $Q$ (3, 2)

and let **v** be represented by the directed line segment from

R (1, 2) to S (4, 4)

Show that **u** = **v**.

## Solution :

From the Distance Formula, it follows that $\overrightarrow{PQ}$ and $\overrightarrow{RS}$ have the same magnitude.

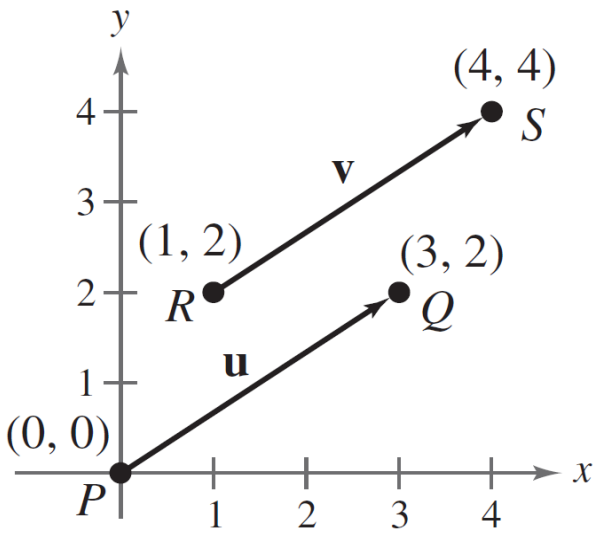
$$\|\overrightarrow{PQ}\| = \sqrt{(3-0)^2 + (2-0)^2} = \sqrt{13}$$

$$\|\overrightarrow{RS}\| = \sqrt{(4-1)^2 + (4-2)^2} = \sqrt{13}$$

Moreover, both line segments have the same direction, because they are both directed toward the upper right on lines having the same slope.

$$\text{Slope of } \overrightarrow{PQ} = \frac{2-0}{3-0} = \frac{2}{3}$$

$$\text{Slope of } \overrightarrow{RS} = \frac{4-2}{4-1} = \frac{2}{3}$$




So, $\overrightarrow{PQ}$ and $\overrightarrow{RS}$ have the same magnitude and direction, and it follows that **u** = **v**.

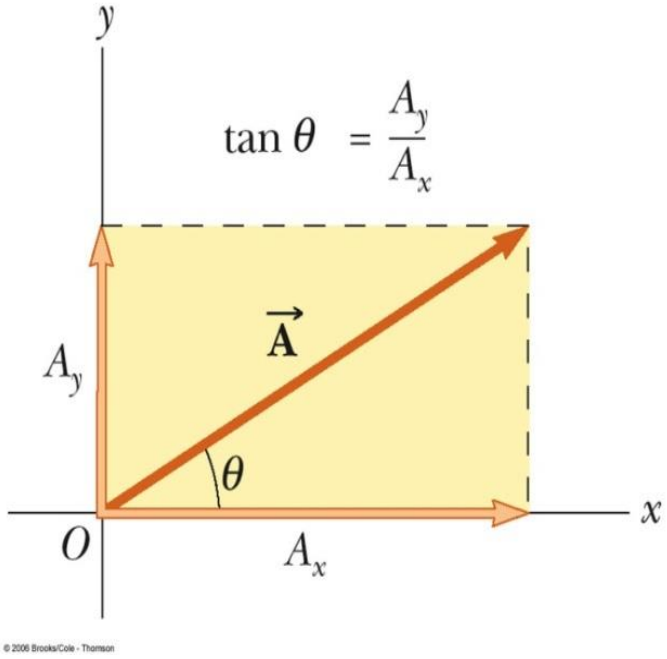
# From Polar to Cartesian coordinates and vice versa

- The x-component of a vector is the projection along the x-axis

$$\cos\theta = \frac{A_x}{A} \qquad A_x = A\cos\theta$$

- The y-component of a vector is the projection along the y-axis

$$\sin\theta = \frac{A_y}{A} \qquad A_y = A\sin\theta$$




- Then, the direction $\tan\theta = \frac{A_y}{A_x}$ $\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$
- And the magnitude

$$|A| = \sqrt{{A_x}^2 + {A_y}^2}$$

## *Example*

*Find the Cartesian form of* $\vec{A} = 55 \langle 205^o$

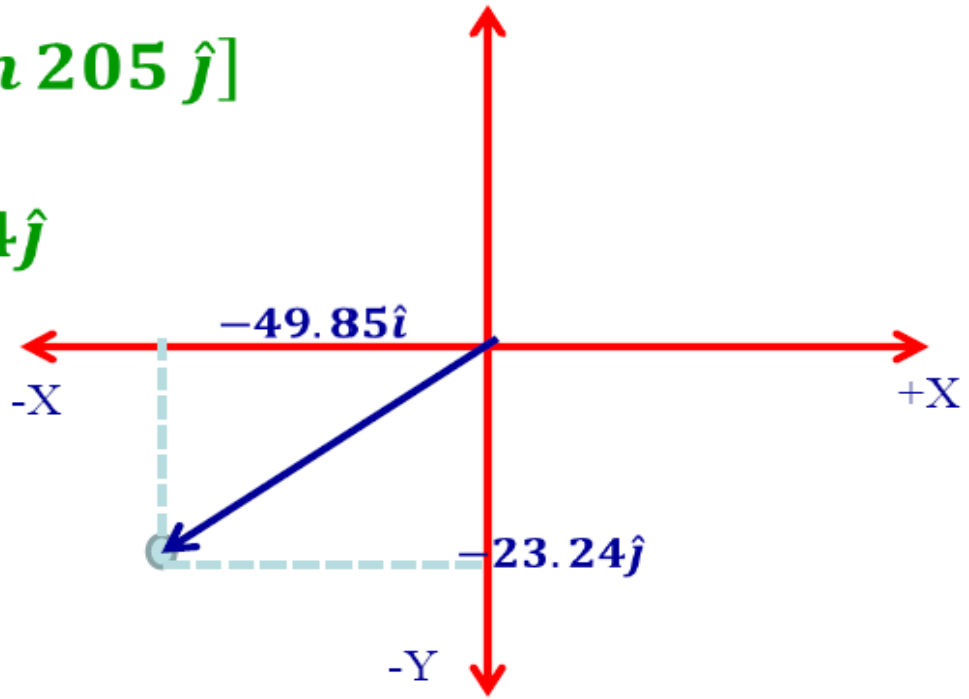
## *Solution*

***In general :***

$$\vec{A} = |A| \, [cos\,\theta \, \hat{\imath} + sin\,\theta \, \hat{\jmath}]$$

$$\vec{A} = 55 \, [cos\,205 \, \hat{\imath} + sin\,205 \, \hat{\jmath}]$$

$$\vec{A} = -49.85\hat{\imath} \; - 23.24\hat{\jmath}$$



# *Example*

***Find the polar form of*** $\vec{A} = -3\hat{\imath} + 4\hat{\jmath}$,

## *Solution*

$$A = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + (4)^2} = 5$$

***To find the angle by*** ⟹ $\tan\theta = \frac{y}{x}$

$$\tan\theta = \frac{4}{-3} \Longrightarrow \theta = \begin{cases} -53.13^o & \times \\ 126.87^o & \surd \end{cases}$$

***The vector in polar form*** ⟹ $\vec{A} = 5\langle 126.87^o$

# Unit Vectors

A unit vector is simply a vector of unit length with the same direction as a given vector.

Rule

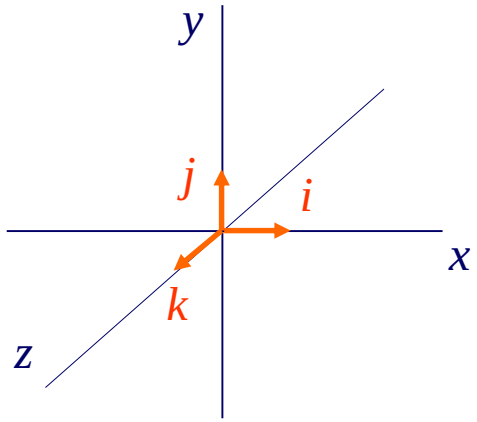
$$\hat{n} = \frac{\vec{A}}{|A|}$$

which has unit magnitude and is in the direction of $\vec{A}$ .

The unit vectors in the positive direction of x, y and z axis are labeled as $\hat{i}$ , $\hat{j}$ and $\hat{k}$ respectively with a small hat on top.

Then we can express the vector $\vec{A}$ as,

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$



## *Example*

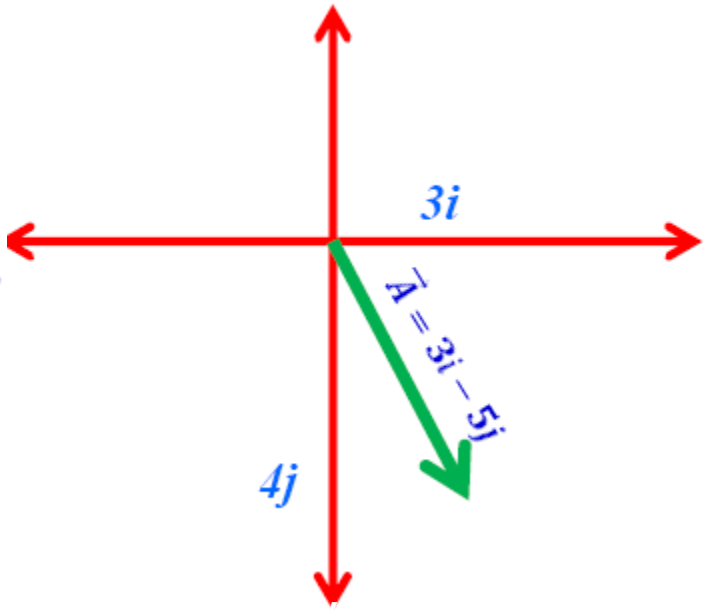
- **Find the unit vector of the following Cartesian vector A**

$$\vec{A} = 3i - 5j$$

## *Solution*

$$\hat{A} = \frac{\vec{A}}{|A|}$$

$$\hat{A} = \frac{3i - 5j}{\sqrt{9 + 25}} = \frac{3i - 5j}{5.83} = 0.515i - 0.858j$$



# 5- Vector Multiplication

## * Cross Product

The cross product **a** × **b** (read "a cross b"), is a vector that is perpendicular to both **a** and **b**, and thus normal to the plane containing them.

*definition*

$$\vec{a} \times \vec{b} = |\vec{a}|\,|\vec{b}| \sin\theta\, \hat{n}$$

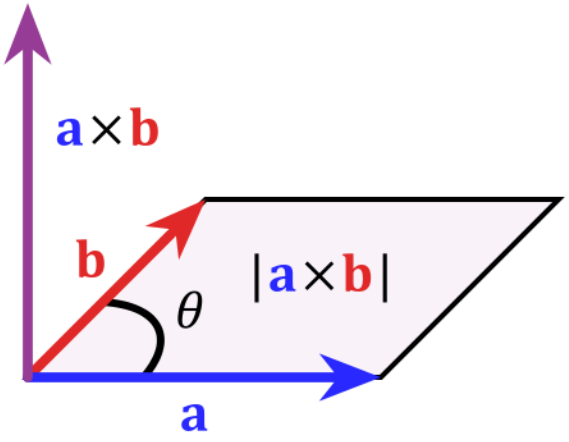
$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} \qquad \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$$

$$\vec{a} \times \vec{b} = i\,(a_2b_3 - a_3b_2) - j\,(a_1b_3 - a_3b_1) + k\,(a_1b_2 - a_2b_1)$$

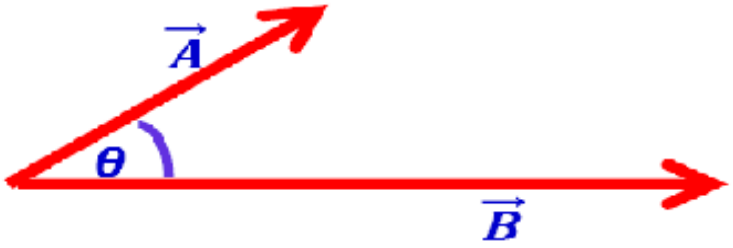
$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$i \times j = k$ $\quad j \times k = i$ $\quad k \times i = j$

$j \times i = -k$ $\quad k \times j = -i$ $\quad i \times k = -j$



# Dot Product




*definition* $\vec{A}.\vec{B} = |A||B| \cos\theta$

*If* $\vec{A} = x_1\hat{\imath} + y_1\hat{\jmath} + z_1\hat{k}$

$\vec{B} = x_2\hat{\imath} + y_2\hat{\jmath} + z_2\hat{k}$

*Rule* $\vec{A}.\vec{B} = x_1x_2 + y_1y_2 + z_1z_2$

*As we know* $\hat{\imath}.\hat{\imath} = \hat{\jmath}.\hat{\jmath} = \hat{k}.\hat{k} = 1$

*and* $\hat{\imath}.\hat{\jmath} = \hat{\jmath}.\hat{k} = \hat{k}.\hat{\imath} = 0$

## *Example*

*Find the angle between* $\vec{A} = -10\hat{\imath} - 25\hat{\jmath} - 50\hat{k}$ *and* $\vec{B} = -12\hat{\imath} + 20\hat{\jmath}$

## *Solution*

*rule* $\quad \vec{A}.\vec{B} = |A||B| \cos\theta$

*since* $\quad \cos\theta = \dfrac{\vec{A}.\vec{B}}{|A||B|}$

*then* $\quad \cos\theta = \dfrac{(-10\hat{\imath} - 25\hat{\jmath} - 50\hat{k}).(-12\hat{\imath} + 20\hat{\jmath})}{\sqrt{100 + 625 + 2500}\sqrt{144 + 400}}$

*then* $\quad \cos\theta = \dfrac{(120 - 500)}{(56.8)(23.3)} = \dfrac{-380}{1324.8} = -0.2868$

*then* $\quad \theta = 106.67^{o}$